Asynchronous Event-Based Spectroscopy for Microsecond-Scale Spectral Reconstruction

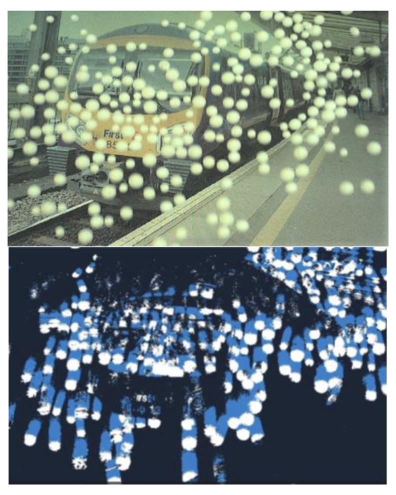
Joana Teixeira, Tomás Lopes, Pedro A.S. Jorge, Nuno A. Silva

INESC TEC, Centre of Applied Photonics, Rua do Campo Alegre 687, 4169-007 Porto, Portugal

Abstract. Many physical and chemical processes evolve on timescales over the limits of conventional instrumentation. Here, we present an event-based spectrometer that enables spectral reconstruction with microsecond resolution through asynchronous sensing. We employ an event-based vision sensor to record changes in the dispersed optical signal, which are then processed into a spectral signal. Our approach can reconstruct spectra at effective probing rates of up to tens of kHz, exceeding the practical limits of conventional middle-range spectrometers.

Towards a μ-second resolution neuromorphic spectrometer

Event-based sensors. Fully asynchronous pixels output a stream of +1/-1 "events" according to the increase/decrease of light reaching the detector.



Frame-Based

Event-Based

- Static scenes aren't registered;
- Dynamic Range >120 dB;
- Pixel latency <100 μs;
- Low-cost alternative to high-speed sensors.

Event-based spectrometer (EVS). Czerny-turner configuration spectrometer where the light is directed to a HD event-based sensor to a reference frame-based camera.

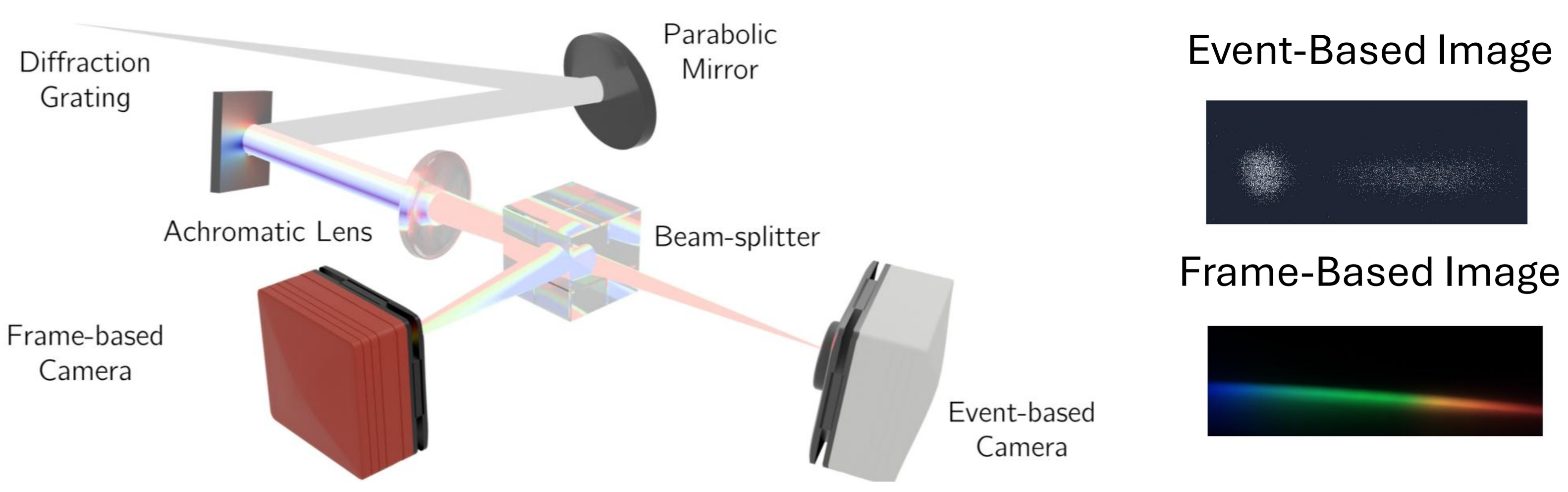


Figure 1. Illustration of the experimental setup and the acquisitions resulting from the frame-based and event-based camera. The same signals are also sent to a conventional spectrometer for calibration and comparison.

Signal Processing Pipeline. The event-stream is broken down into "frames" by accumulating each pixel's events over a certain Δt (defined in post-processing).

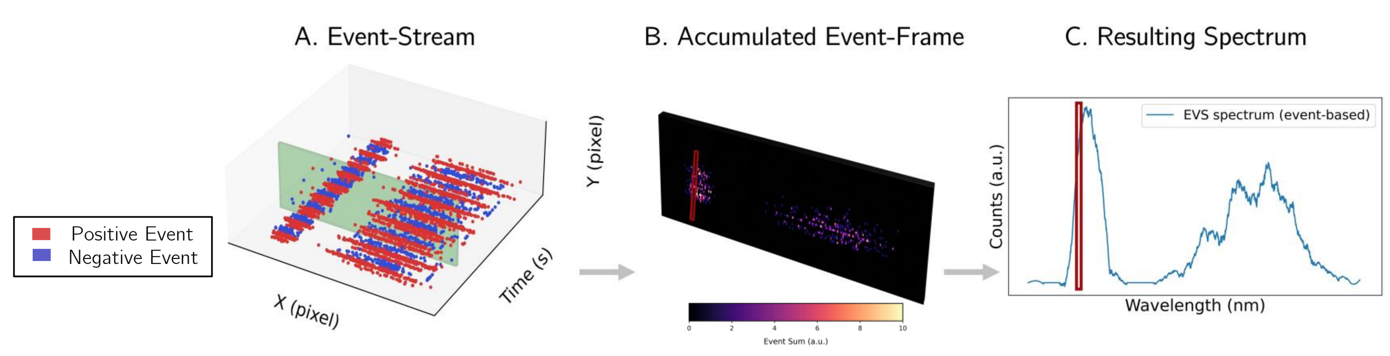


Figure 2. The event "frames" are transformed into spectra by summing along the columns. To reduce the noise impact a subset of rows centered on the "higher intensity row" are selected. The spectra are smoothed by a Savitzky–Golay filter.

Key Takeaways and Perspectives

1. **Enhanced Contrast**: Inherent suppression of constant illumination improves spectral feature identification in low-light conditions;
2. **Low-Latency Dynamic Tracking**: Successfully resolved the progressive attenuation of spectral peaks during microfluidic dye injection, capturing intermediate states invisible to standard sensors;
3. **Efficient Data Handling**: Conditional reconstruction eliminates redundant data during static periods, significantly reducing computational and storage overhead;
4. **High-Speed Temporal Resolution:** Under the right illumination, an effective sampling rate of 80 kHz was achieved, vastly outperforming the conventional frame-based reference.
5. **Microfluidics Integration**: Monitoring of transient chemical reactions and high-speed droplet sorting by bypassing the frame-rate bottleneck, allowing for spectral updates at the micro-second scale.

Experimental Results

Case 1. Spectral Calibration

Modulated
Red LED
Blue LED

Conventional and event-based spectrometer

Linear wavelength–pixel mapping

- 234nm bandwidth
- 0.18nm spectral resolution

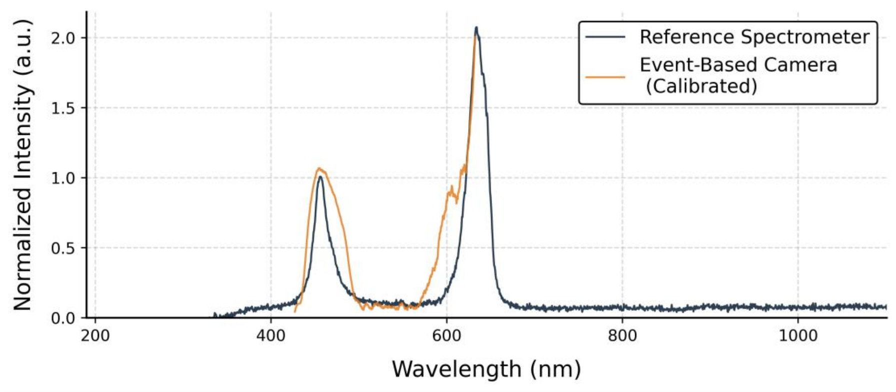


Figure 3. Spectral Calibration: Spectrometer vs Event-based camera

Case 2. Modulated Illumination Reconstruction

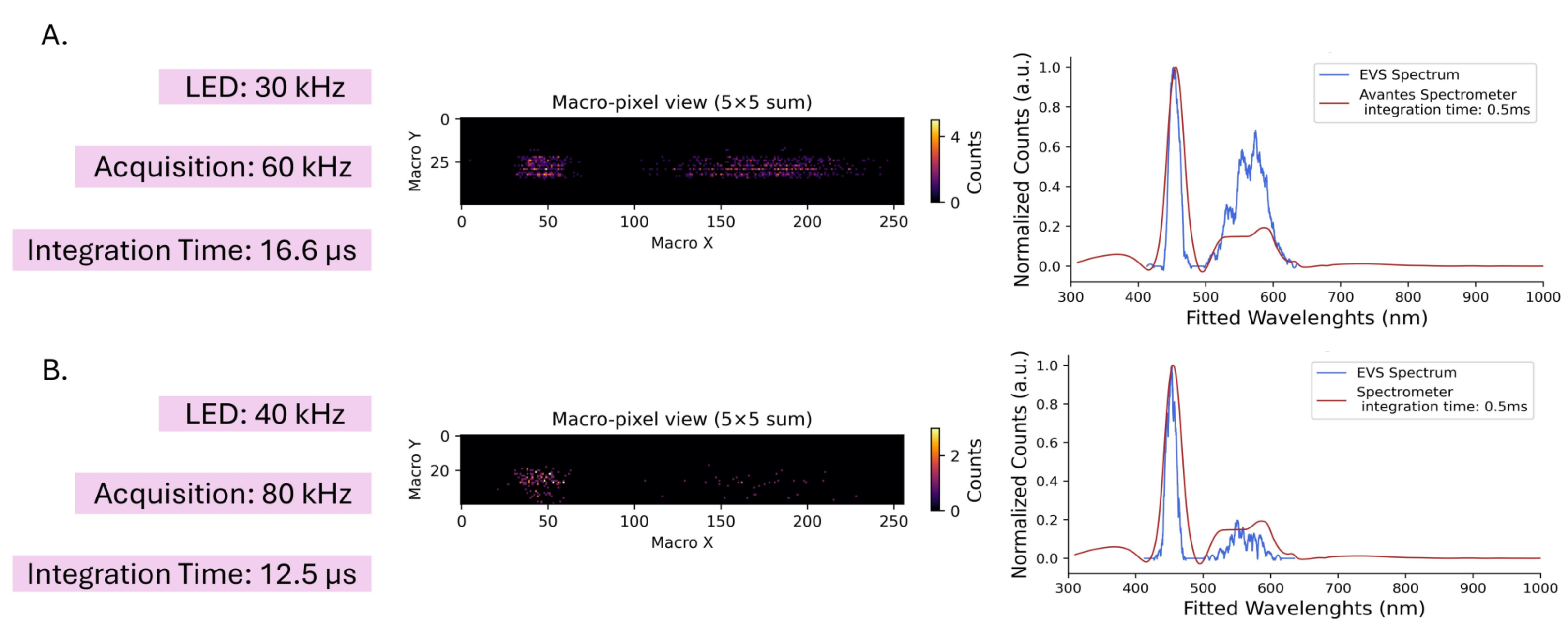


Figure 4. Accumulated event image and resulting spectra for a white LED modulated at 30 and 40 kHz. Our method allows for **spectral reconstruction up to 80kHz**, against the conventional spectrometer which has a maximum acquisition rate of 1kHz.

Case 3. Study of Samples in a Microfluidics Environment

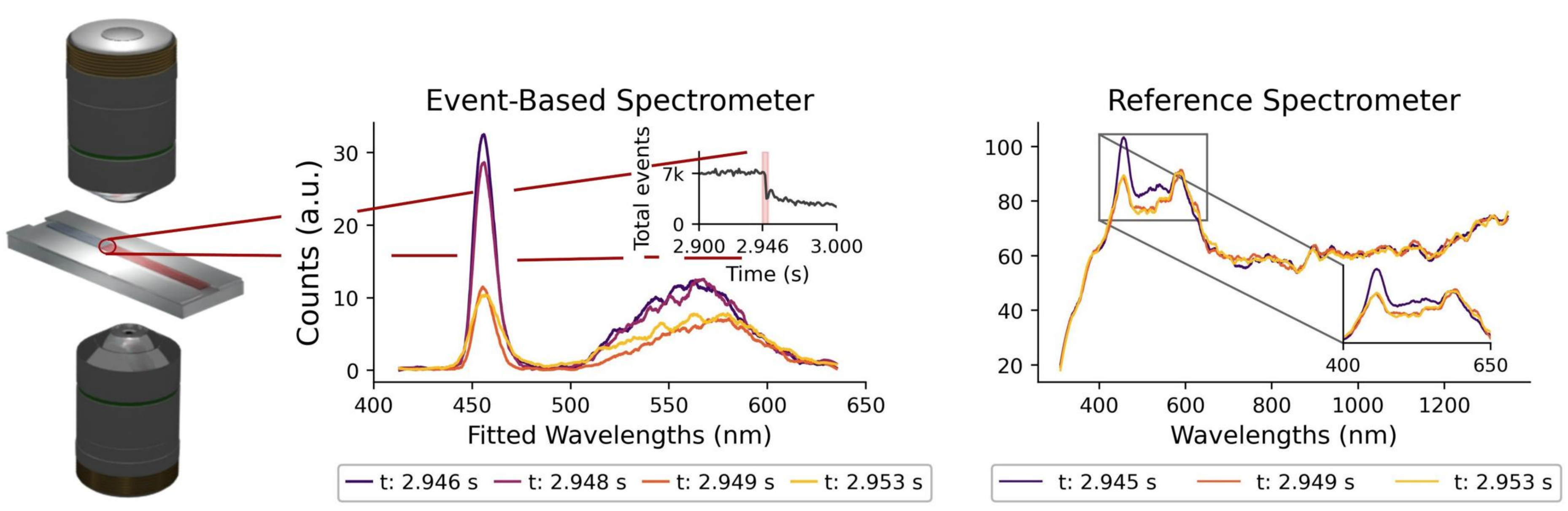
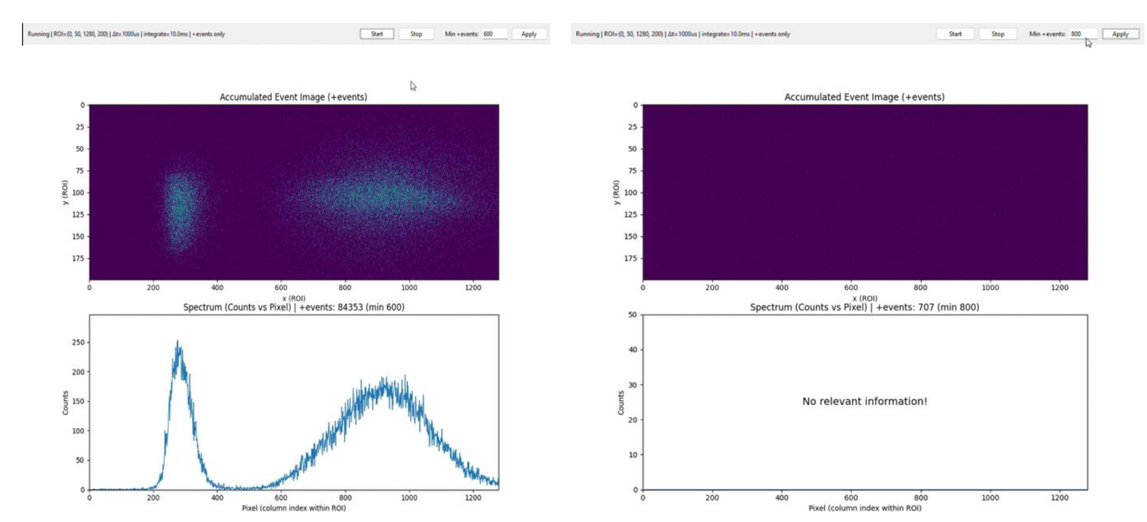


Figure 5. Real-time spectral tracking of dynamic microfluidic flow, during the introduction of red food dye. The EVS demonstrates superior temporal resolution, capturing **multiple intermediate spectral transitions** as the dye attenuates the blue peak. The EVS also provides enhanced spectral contrast by **inherently filtering constant background illumination.**

Case 4. Real-Time Operation

Figure 6. High-speed, adaptive spectral monitoring. The data is only processed when activity exceeds a defined threshold, enabling low-latency tracking of dynamic changes **without the redundant data storage.**



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Our team